

BINITA RANA

Computer Engineer | Aspiring Data Scientist | Machine Learning & Python-Django Developer

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Professional Summary

Computer Engineering graduate with hands-on experience in machine learning, natural language processing, Python/Django development, and application development. Experienced in building and evaluating machine-learning solutions for real-world problems, including healthcare risk prediction and automated fake-news detection. Currently working as a Python Django Intern and preparing to pursue an M.S. in Data Science. Interested in machine learning, data-driven applications, healthcare analytics, and intelligent software systems.

Education

Bachelor of Engineering in Computer Engineering

-Nepal College of Information and Technology (NCIT), Lalitpur, Nepal

Affiliated with Pokhara University

Graduated: November 2025 CGPA: 3.02/4.00

Internship Experience

Python Django Intern

July 2026 - Present

- Developing web applications using Python and Django.
 - Building CRUD (Create, Read, Update, Delete) applications for database-driven systems.
 - Working with Django models, views, templates, and forms.
 - Performing database operations using SQLite.
 - Gaining practical experience in backend development, debugging, and web application deployment.
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Academic Project

HealthRisk AI — Personalized Disease Risk Prediction Platform

- Developed a machine-learning platform for estimating diabetes, heart disease, and stroke risk using health and lifestyle data.
- Performed data preprocessing, exploratory data analysis, feature preparation, model training, and evaluation.
- Experimented with Random Forest, Extra Trees, Gradient Boosting, and AdaBoost classification models.
- Performed model tuning and evaluation to identify more effective configurations for disease-risk prediction.
- Integrated machine-learning models into a Django web application with health assessments, prediction history, risk visualization, and personalized results.

- Built an interactive dashboard with health history and downloadable health reports.
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AI Fake News Detection System

- Built a machine learning system that classified news articles as real or fake with approximately 95% classification accuracy, trained and evaluated on a publicly available fake news dataset.
 - Implemented text preprocessing, TF-IDF feature extraction, and Random Forest classification while experimenting with BERT.
 - Integrated the prediction model into a Django web application, resolving frontend-backend communication challenges and enabling near real-time predictions through a user-friendly web interface.
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Event Management App

- Designed and developed a Flutter-based event management app used by approximately 80 participants during a college-wide event, replacing a manual registration process.
 - Automated user registration and event management using Firebase-backed workflows.
 - Implemented real-time notifications to keep participants informed about event updates.
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Certifications

- **Flutter Application Development – Broadway Infosys**
December 2023 – January 2024
 - **Python with Django Framework – Kitwosd It Support Center**
April 2025– June 2026
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Extra-Curricular Activities

- Participated in project analysis and collaborative software development activities as part of academic coursework at Pokhara University.
 - Worked in team-based software engineering projects involving planning, implementation, testing, and documentation.
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Technical Skills

- Programming: Python, C++, Dart
 - Machine Learning & NLP: Scikit-learn, TF-IDF, Random Forest, BERT (Basic)
 - Frameworks: Django, Flutter
 - Databases: SQLite, Firebase
 - Web Technologies: HTML, CSS, JavaScript, React
 - Tools: Git, GitHub, VS Code
 - Core Concepts: Machine Learning, Data Structures & Algorithms, Object-Oriented Programming
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Relevant Coursework

- Data Structures and Algorithms
 - Database Management Systems
 - Artificial Intelligence
 - Machine Learning
 - Software Engineering
 - Computer Networks
 - Operating Systems
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